Blindspot Stellar Spectra Denoiser

Motivation

Background:

- **Challenge of Noise:** Low signal-to-noise ratio (SNR), especially when SNR per resolution element drops below 3, hampers data quality.

Main Motivation: Real-Time Decision Making

- **Our Methods Runs on extremely low SNR spectra in milliseconds, enabling quick target verification:** After a single exposure, if we detect we're observing the wrong object, we can immediately retarget the fiber.

  Example: Aiming to observe a dwarf galaxy but noticing the radial velocity indicates a Milky Way object—even with large errors—signals a misidentification.

Significance: Rapid identification saves valuable observation time and resources.

Limitations of Current Methods: Template Fitting with Maximum Likelihood

- Slow due to computational complexity.
- Relies on imperfect templates.
- Difficult to find optimal initial values for iterative convergence.
- Breaks down when SNR per resolution element is less than 3.

Objective: Develop a novel denoising method that preserves key spectral features like absorption lines while reducing noise effectively. Facilitate faster and more accurate real-time decisions during observations.

Blindspot Denoiser Architecture

Innovative Approach: Adapted Blindspot Network for 1D Spectral Data

- Originally designed for image denoising without using the corrupted pixel's own value.
- L1 trains much faster than L2,
- Modified to handle the complexities of spectral data.

Key Features:

Unsupervised vs. Supervised Learning:

- **Traditional blind-spot networks are unsupervised.**
- **Unsupervised methods didn't work out-of-the-box for spectra due to the need to capture narrow spectral features.**
- **Implemented a supervised, model-based approach to enhance performance.**

Predicts Data Points Without Self-Referencing:

- **Prevents the network from relying on potentially corrupted values.**

Optimized Loss Function:

- **Emphasizes spectral fidelity to preserve critical features like absorption lines.**

Technical Details:

- **U-Net Architecture with Blind-Spot Convolutional Layers:** Allows the network to learn from surrounding data points without self-contamination.
- **Customized Loss Functions:** Utilizes weighted L2, L1, and leverage score loss functions for improved noise handling.
- **Difference Between Images and Spectra:**
  - - Images:
  - - For human consumption; noise perception is subjective.
  - - Small kernels suffice due to local feature importance.
  - - Spectra:
  - - Contain detailed information analyzed by algorithms (e.g., template fitting).
  - - Require wider kernels to capture line profiles and broader contextual information.

Denoising Results: Demonstrates superior noise reduction, significantly improving SNR and accuracy of stellar parameter inference like radial velocity.

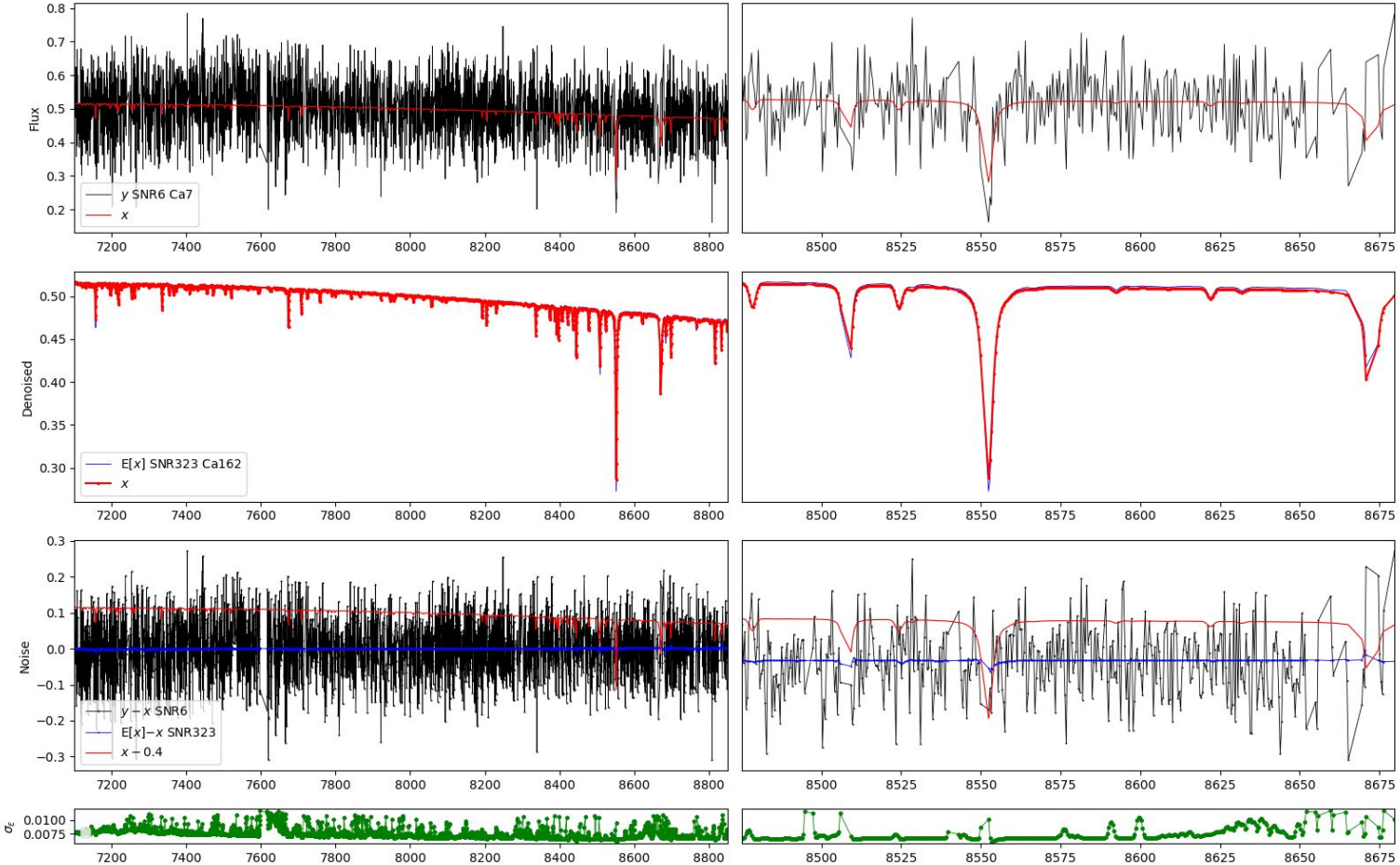
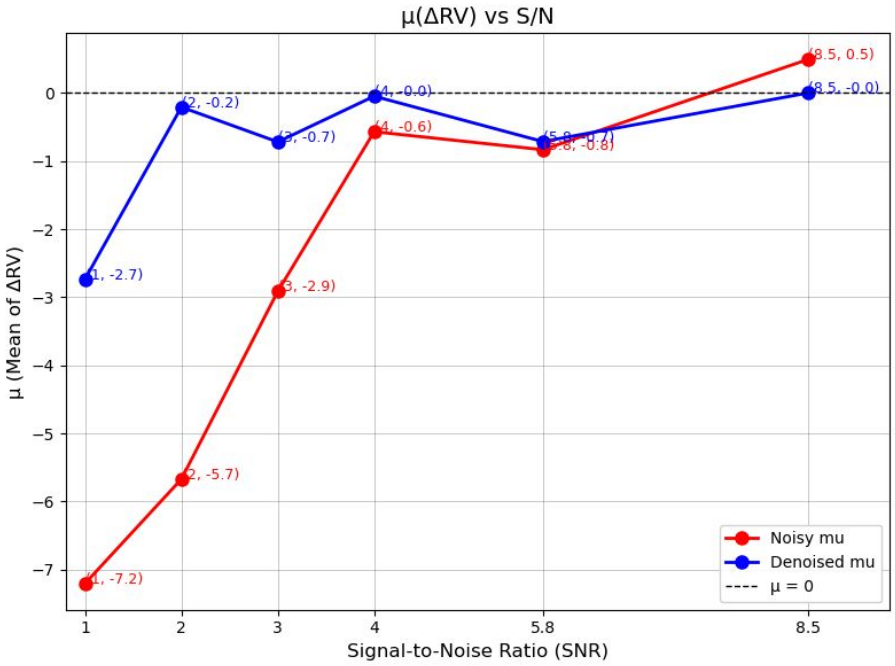
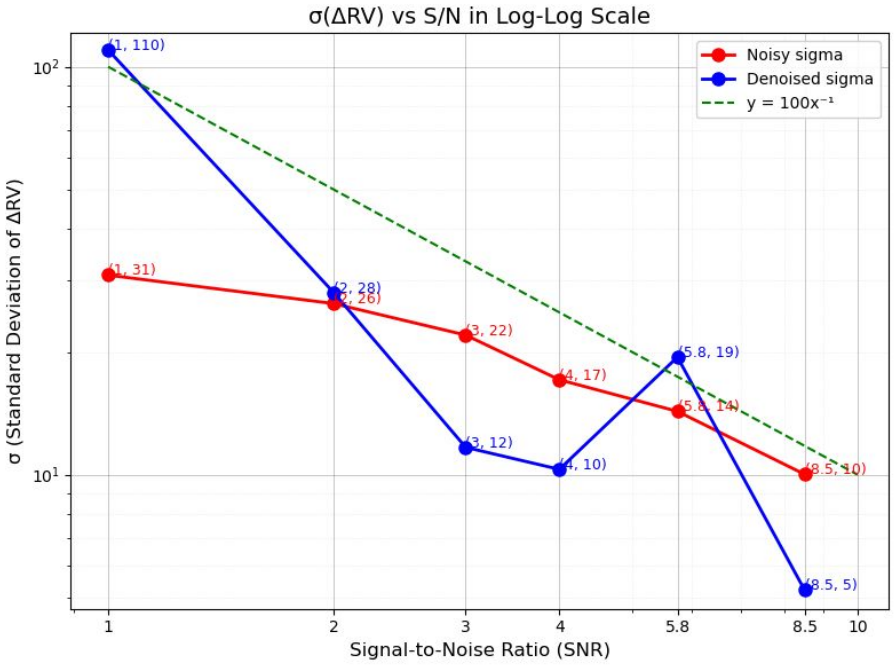
Significantly improves SNR per resolution element.

Maintains critical details like absorption lines.

Improves accuracy in determining parameters such as radial velocity.

Enables quicker assessments during observations, optimizing telescope time.

85% skylines are being masked out.



Superior Radial Velocity Estimations after just 1 exposure